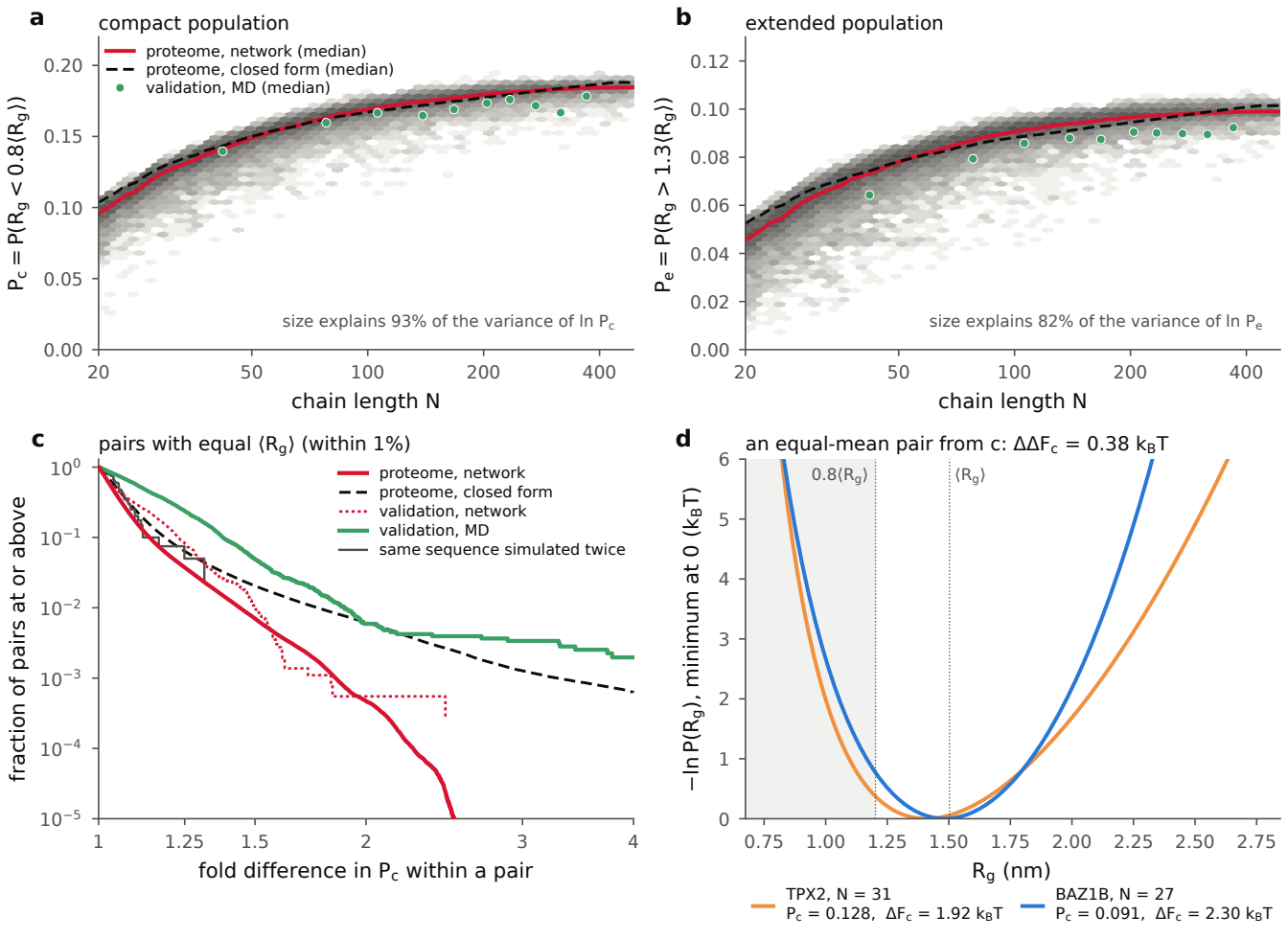

a
compact population
proteome, network (median)
proteome, closed form (median)
validation, MD (median)
$P_c = P(R_g < 0.8\langle R_g \rangle)$
size explains 93% of the variance of ln P_c
chain length N
b
extended population
$P_e = P(R_g > 1.3\langle R_g \rangle)$
size explains 82% of the variance of ln P_e
chain length N
c
pairs with equal $\langle R_g \rangle$ (within 1%)
proteome, network
proteome, closed form
validation, network
validation, MD
same sequence simulated twice
fraction of pairs at or above
fold difference in P_c within a pair
d
an equal-mean pair from c: $\Delta\Delta F_c$ = 0.38 k_BT
$0.8\langle R_g \rangle$
$\langle R_g \rangle$
$-\ln P(R_g)$, minimum at 0 (k_BT)
R_g (nm)
TPX2, N = 31
P_c = 0.128, ΔF_c = 1.92 k_BT
BAZ1B, N = 27
P_c = 0.091, ΔF_c = 2.30 k_BT